

Product Requirements Document (PRD)

Product Name: GramCare AI – Affordable Smart Healthcare for Remote Villages

1. Problem Statement

Remote villages lack access to hospitals and doctors. Residents must travel long distances for treatment, leading to delayed diagnosis, high costs, untreated illnesses, and health risks. The solution must improve healthcare accessibility and delivery using affordable technology suitable for low-infrastructure environments.

2. Product Vision

To build a low-cost AI-powered healthcare platform that enables remote diagnosis, virtual doctor consultations, health record management, and emergency support for rural communities.

3. Target Users

Primary Users:

- Rural villagers
- Community Health Workers (CHWs)
- Government rural clinics

Secondary Users:

- Urban doctors
- NGOs
- Public health administrators

4. Core Features (MVP)

- AI Symptom Checker with multilingual support
- Telemedicine (low-bandwidth audio/video)
- Digital Health Records with Patient ID

- Emergency Alert System with location sharing
- Community Health Worker Dashboard
- Offline Mode with sync functionality

5. AI Components

- Symptom-to-Diagnosis NLP model
- Risk Scoring Engine (Low/Medium/High)
- Voice-to-Text regional language support

6. Technical Architecture

Frontend: Flutter or React Native

Backend: Node.js or FastAPI

Database: Firebase / Supabase / SQLite (offline)

AI Layer: Lightweight LLM API + Rule-based triage engine

Cloud: AWS/GCP Free Tier

7. Non-Functional Requirements

- App size under 50MB
- Works on Android 8+
- Offline-first architecture
- End-to-end encryption
- Role-based access control

8. Monetization Model

- Government partnerships
- NGO collaborations
- Low consultation fee (₹10–₹30)
- Clinic subscription plans

9. Success Metrics (KPIs)

- Rural user adoption rate

- Reduction in travel-based visits
- AI triage accuracy
- Doctor consultation completion rate
- Improvement in maternal and child health tracking

10. Future Enhancements

- IoT health kit integration
- Drone medicine delivery
- Predictive outbreak detection
- Integration with national healthcare schemes